Light-induced Phase Transitions and Nonlinear Responses in 1D/2D Topological Insulators

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Can a laser field modify the topology of a topological insulator?

Models studied: SSH · Rice-Mele (1D) | QWZ · Haldane (2D)

Part I	Part II	Part III
Topological Band Theory	Floquet Theory & Light-Matter Coupling	High Harmonic Generation

Part I

Topological Band Theory

Bloch's Theorem & Crystal Momentum

Bloch's Theorem

In a periodic potential $V(\mathbf{r}+\mathbf{a}) = V(\mathbf{r})$:

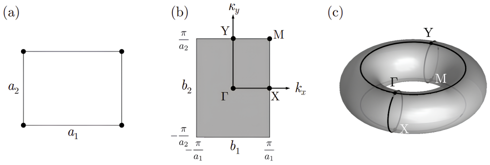
$$|\psi_{n\mathbf{k}}\rangle = e^{i\mathbf{k}\cdot\mathbf{r}}|u_{n\mathbf{k}}\rangle$$

$|u_{n\mathbf{k}}\rangle$ is *cell-periodic*; $\mathbf{k}$ is crystal momentum.

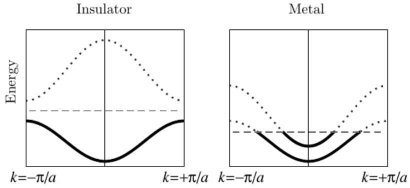
- $\mathbf{k}$ is defined only modulo a reciprocal lattice vector $\mathbf{G}$

$\Rightarrow$ Restricted to the **Brillouin zone** (BZ)

- The BZ has the topology of a torus T^d
- Band structure: $E_n(\mathbf{k})$ from $H(\mathbf{k})|u_{n\mathbf{k}}\rangle = E_n(\mathbf{k})|u_{n\mathbf{k}}\rangle$



BZ topology: rolling the rectangular BZ gives a torus



Band diagrams of metals vs insulators

Tight-Binding Model

Hamiltonian

For a 1D chain with nearest-neighbour hopping:

$$H = E_0 \sum_n |n\rangle \langle n| - t \sum_n \left(|n\rangle \langle n+1| + |n+1\rangle \langle n| \right),$$

where t is the hopping amplitude and E_0 is the on-site energy.

Core Assumption

The tight-binding model treats electrons as localized around atomic sites, with hopping between neighbours. Crucially, it **ignores electron-electron repulsion forces** – the electrons are considered independent (single-particle picture).

Berry Phase

Under adiabatic evolution along a closed loop C in the Brillouin zone:

$$\gamma_n = \oint_C \mathcal{A}_n(\mathbf{k}) \cdot d\mathbf{k},$$

where the **Berry connection** is

$$\mathcal{A}_n(\mathbf{k}) = i\langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} | u_{n\mathbf{k}} \rangle.$$

Berry Curvature

The "magnetic field" in momentum space:

$$\Omega_n(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathcal{A}_n(\mathbf{k}),$$

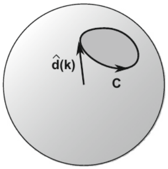
Two-Band Model

For a two-band Hamiltonian:

$$H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma},$$

the Berry phase around a closed loop is half the solid angle subtended by $\hat{\mathbf{d}}$:

$$\gamma_C = \frac{1}{2}\,\Omega_{\text{solid}}.$$



$\hat{\mathbf{d}}(k)$ sweeping the Bloch sphere.

Discrete Symmetries & Topological Invariants

Discrete Symmetries

- **Inversion** $\mathcal{P}$: $\mathcal{P}H(\mathbf{k})\mathcal{P}^{-1} = H(-\mathbf{k})$
- **Time-Reversal** $\mathcal{T}$: $H(\mathbf{k}) = UH(-\mathbf{k})^*U^\dagger$
- **Chiral** Γ: $\Gamma H(\mathbf{k})\Gamma^{-1} = -H(\mathbf{k})$

1D Topological Invariants

- **Zak phase** $\gamma_{\mathrm{Zak}} = \int_{\mathrm{BZ}} \mathcal{A}(k)\, dk \in \{0, \pi\}$
  Protected by inversion *or* chiral symmetry
- **Winding number**
  $\nu = \frac{1}{2\pi} \oint dk\, \frac{\partial}{\partial k} \arg(d_x + id_y)$
  Protected by chiral symmetry

2D Topological Invariants

- **Chern number** $\mathcal{C} = \frac{1}{2\pi} \int_{\mathrm{BZ}} \Omega(\mathbf{k})\, d^2k \in \mathbb{Z}$
  Requires broken time-reversal symmetry (or no TRS)
- $\mathbb{Z}_2$ **invariant** $(-1)^\nu = \prod_i \prod_m \xi_{2m}(\mathbf{\Lambda}_i)$
  Protected by time-reversal symmetry with $\mathcal{T}^2 = -1$

Key property: Invariants are *integers* — they can only change when the **bulk gap closes**

Bulk–Boundary Correspondence

The Central Principle

A nontrivial topological invariant in the bulk **guarantees** gapless states at the boundary — independently of microscopic surface details.

At an interface between two phases with different invariants, the gap must close and reopen $\Rightarrow$ **protected boundary modes**.

- **1D, Zak $= \pi$:** Zero-energy states at chain ends (SSH edge states), protected by chiral symmetry
- **2D, $\mathcal{C} \neq 0$:** $|\mathcal{C}|$ chiral edge modes per boundary, immune to backscattering
- **2D, $\nu = 1$:** Helical edge modes, protected by TRS

Part II

Floquet Theory & Light-Matter Coupling

Light-Matter Coupling: Peierls Substitution

Minimal Coupling: $\mathbf{p} \to \mathbf{p} - q\mathbf{A}$

In a vector potential $\mathbf{A}(t)$, each hopping acquires a **Peierls phase**:

$$t_{\mathbf{mn}} \;\to\; t_{\mathbf{mn}} \exp\left(-\frac{iq}{\hbar}\int_{\mathbf{r_n}}^{\mathbf{r_m}} \mathbf{A}\cdot d\mathbf{r}\right)$$

For spatially uniform fields over bond $\boldsymbol{\delta}$:

$$t \;\to\; t\, e^{-iA(t)\cdot\boldsymbol{\delta}}$$

Current Operator

Rigorously from minimal coupling:

$$\hat{\mathbf{J}} = -\frac{\partial \hat{H}}{\partial \mathbf{A}}$$

The derivative pulls down the bond vector $\boldsymbol{\delta}$: current $\propto$ hopping rate $\times$ distance.

Floquet's Theorem

Time-Evolution Operator

For a time-dependent Hamiltonian $H(t)$,

$$U(t,t_0) = \mathcal{T}\exp\left(-\frac{i}{\hbar}\int_{t_0}^{t} dt'\, H(t')\right).$$

Floquet Theorem

For $H(t+T) = H(t)$, the evolution operator factorizes:

$$U(t,t_0) = e^{-i\hat{K}_F(t)}\, e^{-iH_F(t-t_0)/\hbar},$$

where

- H_F: time-independent **Floquet Hamiltonian**
- $\hat{K}_F(t)$: periodic **kick operator** (micromotion)

Floquet Modes

The Floquet states are

$$|\psi_\alpha(t)\rangle = e^{-i\varepsilon_\alpha t/\hbar}\,|\phi_\alpha(t)\rangle,$$

with $|\phi_\alpha(t+T)\rangle = |\phi_\alpha(t)\rangle$.

Floquet–Brillouin Zone

Quasienergies are defined modulo $\hbar\omega$:

$$\varepsilon_\alpha \equiv \varepsilon_\alpha + m\hbar\omega, \quad m \in \mathbb{Z}.$$

They are usually restricted to the first zone $\varepsilon \in [-\hbar\omega/2, \hbar\omega/2)$.

Floquet Topological Invariants

Floquet Berry Connection

$$\mathcal{A}_F(\mathbf{k}) = \frac{i}{T}\int_0^T \langle u_F(\mathbf{k},t)|\nabla_{\mathbf{k}}|u_F(\mathbf{k},t)\rangle \, dt,$$

where $|u_F(\mathbf{k},t)\rangle$ are the Floquet modes.

Floquet Winding Number (1D)

From the off-diagonal of the one-cycle propagator $U_F(k)$:

$$\nu_F = \frac{1}{2\pi}\int_{-\pi}^{\pi} dk \, \frac{\partial}{\partial k} \arg[U_F^{01}(k)].$$

Gap closings at $\varepsilon = 0$ or $\varepsilon = \pm\pi/T$ signal topological transitions.
ν_F can be $0, 1, 2, \ldots$ – unlike the static case!

Floquet Chern Number (2D)

Define the Floquet Berry curvature:

$$\Omega_F(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathcal{A}_F(\mathbf{k}).$$

Then the Floquet Chern number is

$$C_F = \frac{1}{2\pi}\int_{\text{BZ}} \Omega_F(\mathbf{k}) \, d^2k \in \mathbb{Z}.$$

$C_F \neq 0$ can occur even when the static Chern number $C_0 = 0$!

Part III

High Harmonic Generation

Nonlinear Optics: From Perturbative to Strong-Field

Goal: Probe Topology Optically

We need field strengths comparable to the **atomic binding field**:

$$E_{\text{at}} \approx 5.14 \times 10^{11} \text{ V/m}$$

At such intensities, perturbative optics breaks down entirely.

Nonlinear Susceptibility

Polarization as power series in field:

$$\mathbf{P}(t) = \epsilon_0 \left(\chi^{(1)} \mathbf{E} + \chi^{(2)} \mathbf{E}^2 + \chi^{(3)} \mathbf{E}^3 + \cdots \right)$$

Each $\chi^{(n)}$ generates harmonics at $n\omega$.

Anharmonic Oscillator Picture

Electron in anharmonic potential:

$$U(x) = \frac{1}{2} m \omega_0^2 x^2 + \frac{1}{3} m C_3 x^3 + \cdots$$

Anharmonicity $C_3 \neq 0 \Rightarrow$ response at $2\omega, 3\omega, \ldots$

Expansion **fails** when $E \sim E_{\text{at}}$: enter HHG regime.

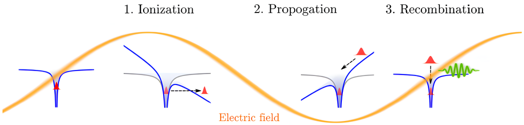
Gas-Phase: The Three-Step Model

Three-Step Mechanism

1. **Tunnel ionization:** intense field suppresses Coulomb barrier, electron freed
2. **Acceleration:** freed electron gains ponderomotive energy $U_p \propto I\lambda^2$
3. **Recollision:** field reverses, electron recombines with ion, emits photon

Cutoff:

$$\hbar\omega_{\max} = I_p + 3.17\,U_p$$



Three-step model: tunnel $\rightarrow$ accelerate $\rightarrow$ recollide

HHG in Solids: Observable & Spectral Features

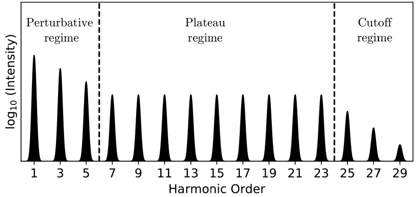
The Basic Observable

Time-dependent macroscopic current $\mathbf{J}(t)$. By Larmor's theorem:

$$S(\omega) \propto \omega^2 \left| \int_{-\infty}^{\infty} \mathbf{J}(t)\, e^{i\omega t}\, dt \right|^2$$

Three Characteristic Regimes

1. **Perturbative** (below gap): $I_{q\omega} \propto I_\omega^q$
2. **Non-perturbative plateau** (above gap): nearly flat intensities
3. **Cutoff**: abrupt drop; scales **linearly** with E_0



Typical solid-state HHG spectrum showing the three regimes

Results

SSH · Rice-Mele · QWZ · Haldane

Laser Parameters

Simulation Parameters

Quantity	Atomic units	Physical units
Frequency ω	0.014 a.u.	≈ 0.38 eV
Wavelength λ	—	$3.2\ \mu$m (mid-IR)
Period T	—	≈ 10.7 fs
Peak field F_0	0.0045 a.u.	$\approx 2.3 \times 10^8$ V/m
Intensity I	—	≈ 0.8 TW/cm^2
Pulse duration	8 cycles	≈ 85 fs

2D: Circular Polarization

$$A_x(t) = \frac{F_0}{\omega}\cos(\omega t), \quad A_y(t) = \frac{F_0}{\omega}\sin(\omega t)$$

Pulse Envelope

Sine-squared: $\sin^2\left(\frac{\pi t}{N_{\text{cyc}} T}\right)$, $N_{\text{cyc}} = 8$

1D: Linear Polarization

$$A(t) = \frac{F_0}{\omega}\cos(\omega t)$$

SSH Model: Static Phases

Bloch Hamiltonian

$$H_{\mathrm{SSH}}(k) = d_x\sigma_x + d_y\sigma_y$$

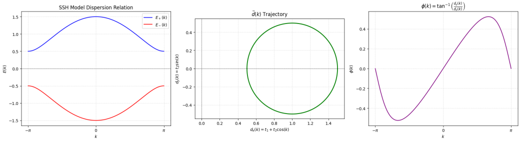
$$d_x = t_1 + t_2\cos k, \quad d_y = t_2\sin k$$

$$E_\pm(k) = \pm\sqrt{t_1^2 + t_2^2 + 2t_1t_2\cos k}$$

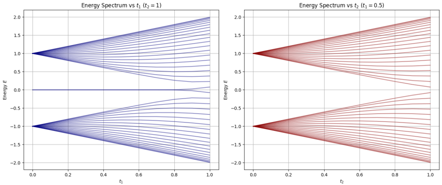
Chiral symmetry: $\Gamma = \sigma_z$, so $d_z = 0$

Topological Phases

- $t_1 > t_2$: $\hat{\mathbf{d}}(k)$ loop **does not** encircle origin $\Rightarrow \nu = 0$ (trivial)
- $t_1 < t_2$: loop **encircles** origin once $\Rightarrow \nu = 1$ (topological)
- $t_1 = t_2$: gap closes, transition



Trivial: $\mathbf{d}(k)$ loop avoids origin

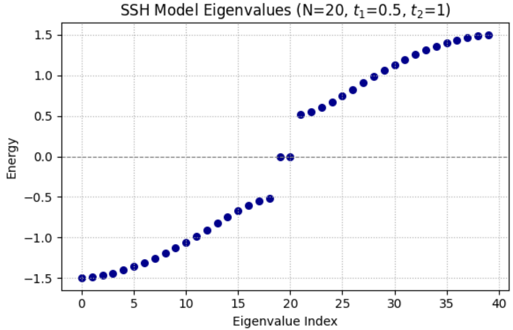


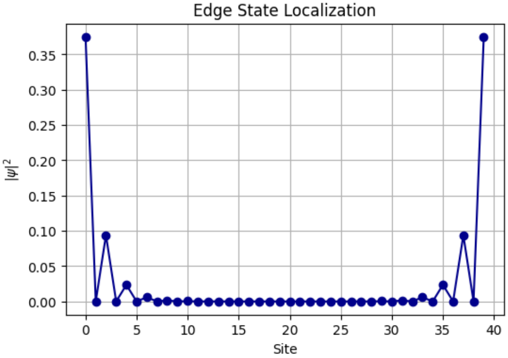
Gap vs hopping: closes at $t_1 = t_2$

SSH: Edge States & Bulk-Boundary Correspondence

Bulk–Boundary Correspondence

$\nu = 1 \Rightarrow$ zero-energy states at chain ends Protected by chiral symmetry Cannot be removed without closing the gap



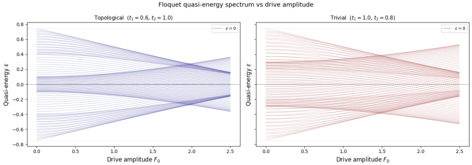


SSH: Floquet Phase Diagram

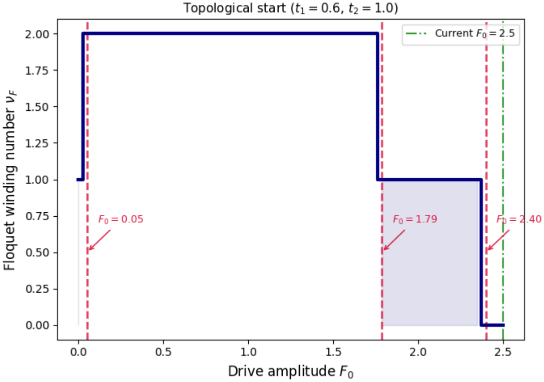
Floquet Winding Number

Drive amplitude F_0 tunes ν_F:

- Gap closings at $\varepsilon = 0$ or $\varepsilon = \pm\pi/T$ signal topological transitions
- ν_F can take values $0, 1, 2, \ldots$

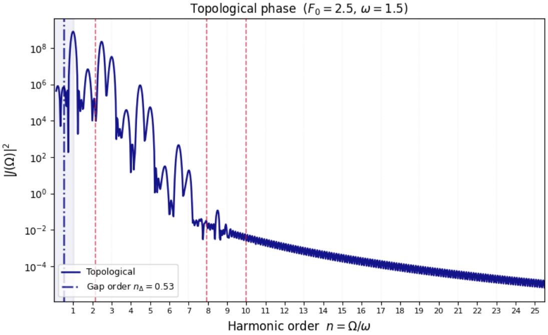


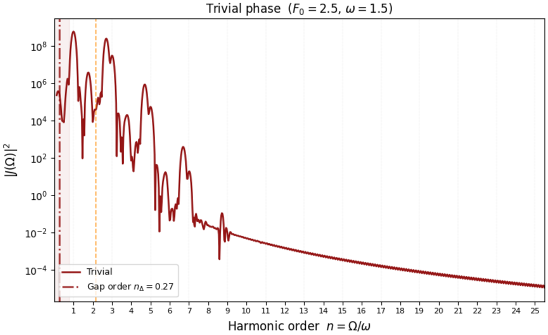
Quasi-energy spectrum vs F_0; gap closings mark transitions



ν_F vs F_0: values $0, 1, 2$ accessible by tuning drive

SSH: HHG Power Spectra





Topological phase ($\nu = 1$): even harmonics clearly visible in plateau

Trivial phase ($\nu = 0$): even harmonics strongly suppressed

SSH: HHG as an All-Optical Topology Probe

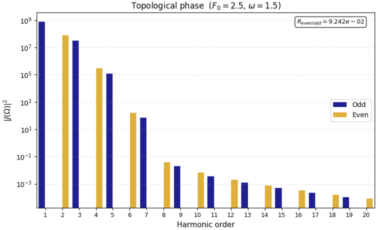
Even-to-Odd Harmonic Ratio

$$R = \frac{\sum_n S_{2n}}{\sum_n S_{2n-1}}$$

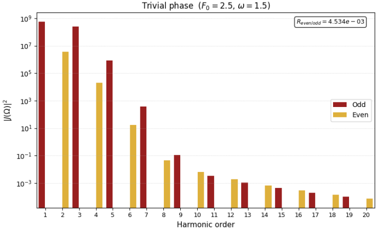
Topological phase ($\nu = 1$): edge states break half-cycle symmetry $J(t + T/2) = -J(t)$ locally
$\Rightarrow$ **even harmonics enhanced**, $R \gg 1$

Trivial phase ($\nu = 0$): half-cycle symmetry preserved
$\Rightarrow$ even harmonics suppressed, $R \approx 0$

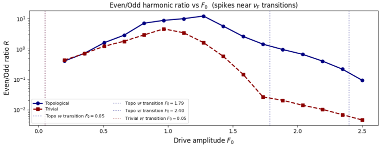
SSH: HHG Spectra & Ratio



Topological phase ($\nu = 1$): significant even-harmonic yield



Trivial phase ($\nu = 0$): even harmonics suppressed



$R(F_0)$: sharp features track Floquet transitions at F_0^*

QWZ Model: Bloch Hamiltonian & Band Structure

Bloch Hamiltonian

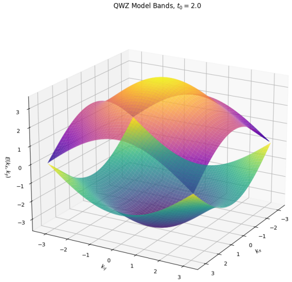
$$\mathbf{d}(\mathbf{k}) = \left(\sin k_x,\ \sin k_y,\ t_0 + \cos k_x + \cos k_y\right)$$

$$H_{\mathrm{QWZ}}(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$$

Gap Closings at High-Symmetry Points

Dirac cones occur when $\mathbf{d} = 0$:

Point	Condition	Chern $\mathcal{C}$ Change
Γ	$t_0 = -2$	$\Delta\mathcal{C} = \pm 1$
X	$t_0 = 0$	$\Delta\mathcal{C} = \pm 1$
M	$t_0 = +2$	$\Delta\mathcal{C} = \pm 1$



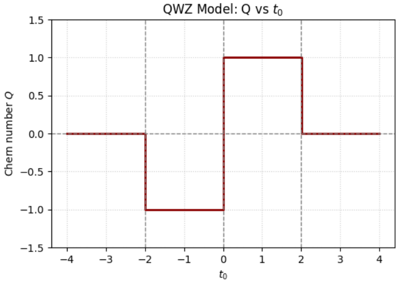
Band structure: conduction (top) and valence (bottom) bands

QWZ Model: Phase Diagram & Chern Number

Phase Diagram

$$\mathcal{C} = \begin{cases} +1 & \text{for } 0 < t_0 < 2 \\ -1 & \text{for } -2 < t_0 < 0 \\ 0 & \text{otherwise (trivial)} \end{cases}$$

- Topological phase: $\mathcal{C} = \pm 1$ with chiral edge modes
- Trivial phase: $\mathcal{C} = 0$, no edge states
- Gap closings at $t_0 = -2, 0, +2$ mark transitions



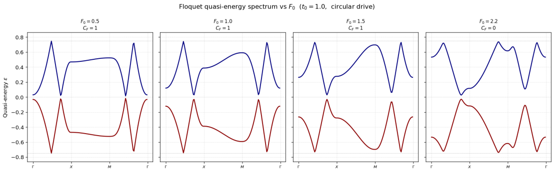
Chern number vs t_0: quantized plateaus

QWZ: Floquet Topological Transitions

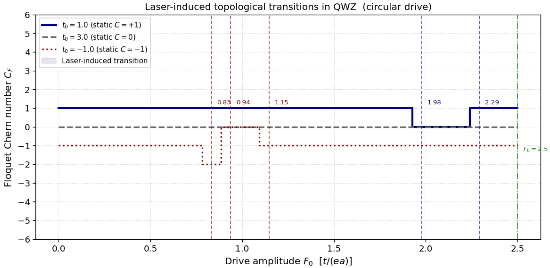
Circularly Polarized Drive

Simultaneously shifts k_x and k_y: entire band structure rotates in BZ.

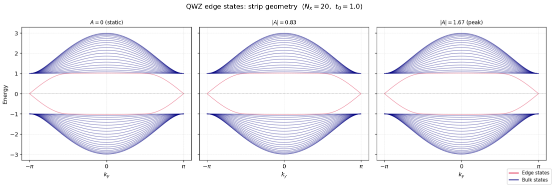
Dynamically breaks TRS $\Rightarrow$ enables $C_F \neq 0$ even from trivial static phase!



Floquet quasi-energy spectrum: gap closings at $\varepsilon = 0$ and $\pm\pi/T$



C_F vs F_0: step-like transitions; $|C_F| > 1$ accessible at large drive



Edge states evolve and annihilate at each topological transition

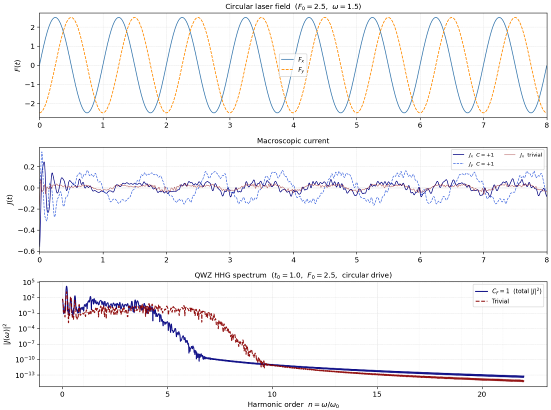
QWZ: HHG Spectra – Topological vs Trivial

Topological vs Trivial Response

- **Topological phase** ($\mathcal{C} \neq 0$): enhanced harmonic yield across the plateau
- **Trivial phase** ($\mathcal{C} = 0$): suppressed emission
- Difference arises from chiral edge mode contributions to the nonlinear current

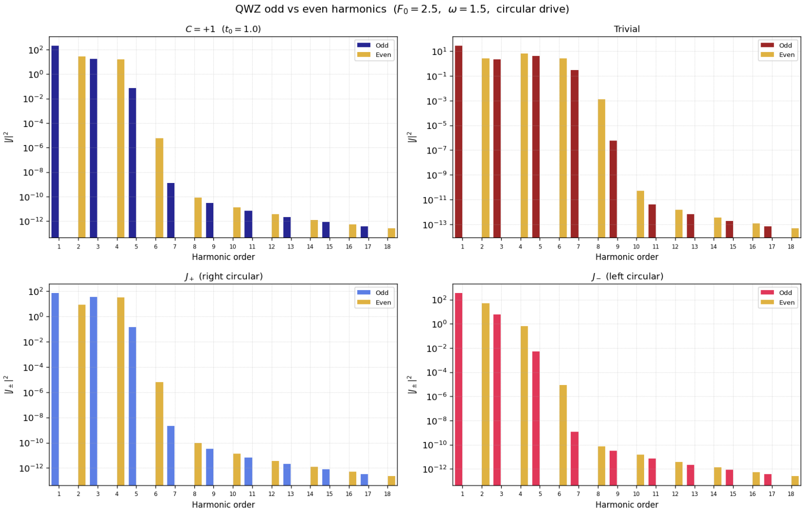
Key Observation

The total HHG spectrum provides a clear, qualitative distinction between topological and trivial phases without requiring helicity resolution.



Total HHG: topological (navy) vs trivial (red)

QWZ: Odd/Even Harmonic Patterns



Distinct harmonic yield patterns encode Chern number

Haldane Model: Chern Insulator on Honeycomb Lattice

Three Ingredients

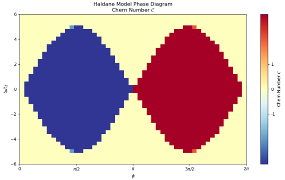
1. NN hopping t_1: two Dirac cones at K and K'
2. **Complex** NNN hopping $t_2 e^{\pm i\phi}$: breaks TRS, opens **topological gap**
3. Staggered potential t_0 (Semenoff mass): breaks inversion, opens **trivial gap**

Bloch Hamiltonian

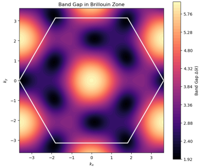
$$H_{\mathrm{H}}(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$$

$$d_x = -t_1 \sum_j \cos(\mathbf{k} \cdot \mathbf{a}_j), \quad d_y = -t_1 \sum_j \sin(\mathbf{k} \cdot \mathbf{a}_j)$$

$$d_z = t_0 - 2t_2 \sin\phi \sum_j \sin(\mathbf{k} \cdot \mathbf{b}_j)$$



Phase diagram: $\mathcal{C} = \pm 1$ lobes in $(\phi,\, t_0/t_2)$ plane

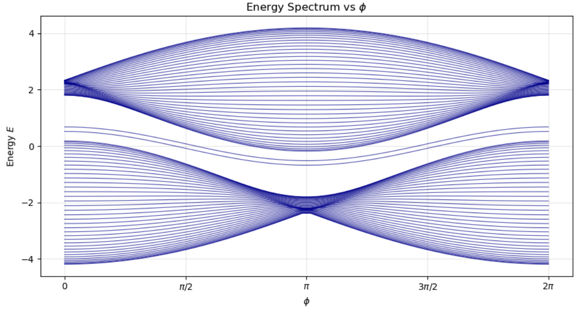


Gap closes only at K and K'

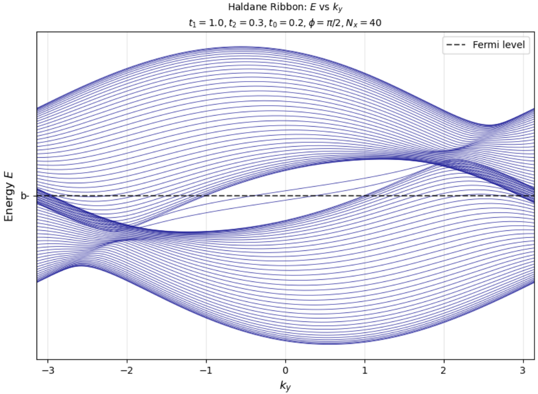
Haldane: Bulk–Boundary Correspondence

Chiral Edge Modes

Unlike the helical modes of $\mathbb{Z}_2$ insulators, Haldane edge states are **strictly unidirectional**: no time-reversal partner exists to allow backscattering.



Spectrum vs ϕ: edge states annihilate at transition



Chiral edge mode in ribbon geometry: one per edge traversing the bulk gap

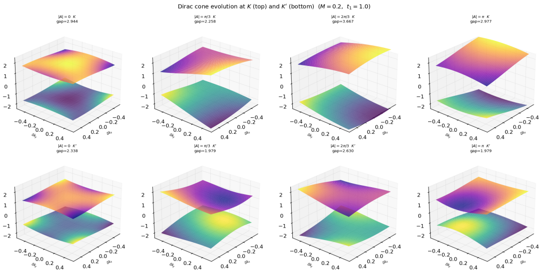
Haldane: Valley-Selective Floquet Engineering

Valley-Selective Response

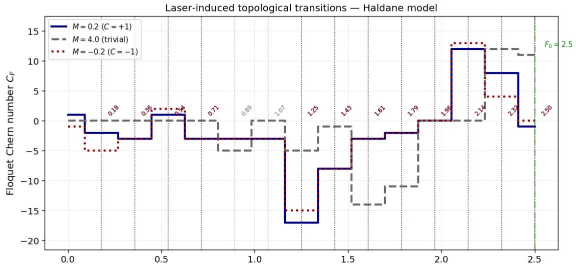
Circular drive addresses K and K' **asymmetrically**:

$$d_z(K,t) \neq d_z(K',t)$$

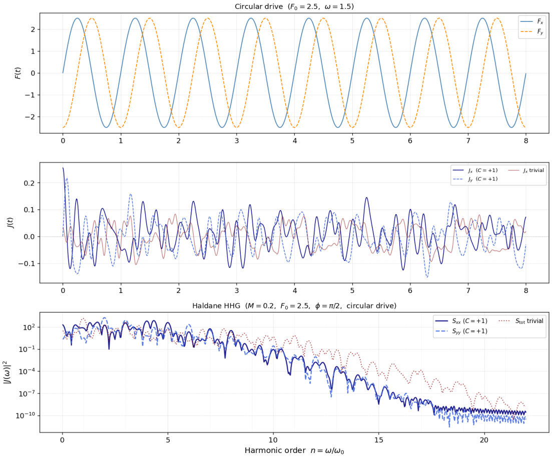
Gap closes at K and K' at **different** critical amplitudes F_0^*!



Dirac cones at K (top) and K' (bottom): asymmetric evolution under circular drive

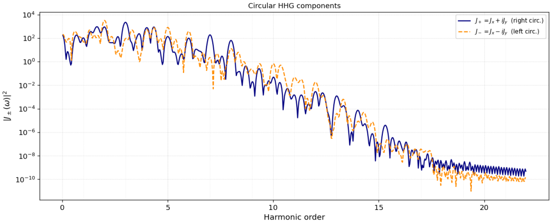


Haldane: HHG Spectrum Under Circular Drive



Total HHG spectrum: topological ($C = +1$, navy) vs trivial (red).

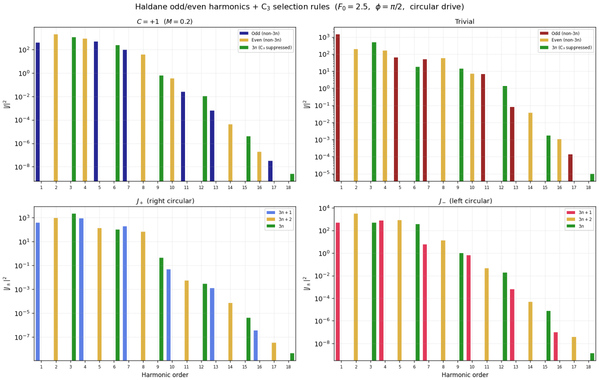
C_3 symmetry suppresses harmonics at orders $3, 6, 9, \ldots$



Helicity decomposition: J_+ (navy) carries $3n+1$ harmonics;

J_- (orange) carries $3n+2$ harmonics

Haldane: Full Harmonic Decomposition



Top row: Total spectrum for topological (left) and trivial (right) phases. **Bottom row:** J_+ (left) and J_- (right) broken into $3n{+}1$, $3n{+}2$, and $3n$ groups.

Green bars: C_3-suppressed $3n$ harmonics.

What the decomposition tells us

- **Topological phase** ($C_F \neq 0$): distinct yield pattern in both J_+ and J_-
- **Trivial phase** ($C_F = 0$): different relative intensities
- $3n$ **harmonics** suppressed in both phases by the lattice symmetry

Conclusions

Yes — lasers can modify topology

- Floquet winding number $\nu_F \in \{0, 1, 2\}$ in SSH tunable by F_0
- Floquet Chern number C_F in QWZ and Haldane passes through multiple integers, including $|C_F| > 1$
- Valley-selective gap closing in Haldane: unique to circular driving
- All transitions are genuinely non-equilibrium — no static counterpart

Limitations & Challenges

- **Numerical solvers**
- **Relaxation Dynamics**
- **Heat Dissipation Effects**

Code: `github.com/Hazem-74/Topological-Insulator-Models`

Thank You